

Ex.No:1: Simulation of FDMA in MATLAB

Aim: To simulate a Frequency Division Multiple Access (FDMA) system using MATLAB and analyze its communication performance in terms of channel bandwidth, data rate, and signal-to-noise ratio (SNR).

Objective 1:

To simulate an FDMA system by allocating **individual frequency channels** to multiple users and determine the **channel bandwidth** assigned to each user.

Output Parameter: Channel Bandwidth (kHz)

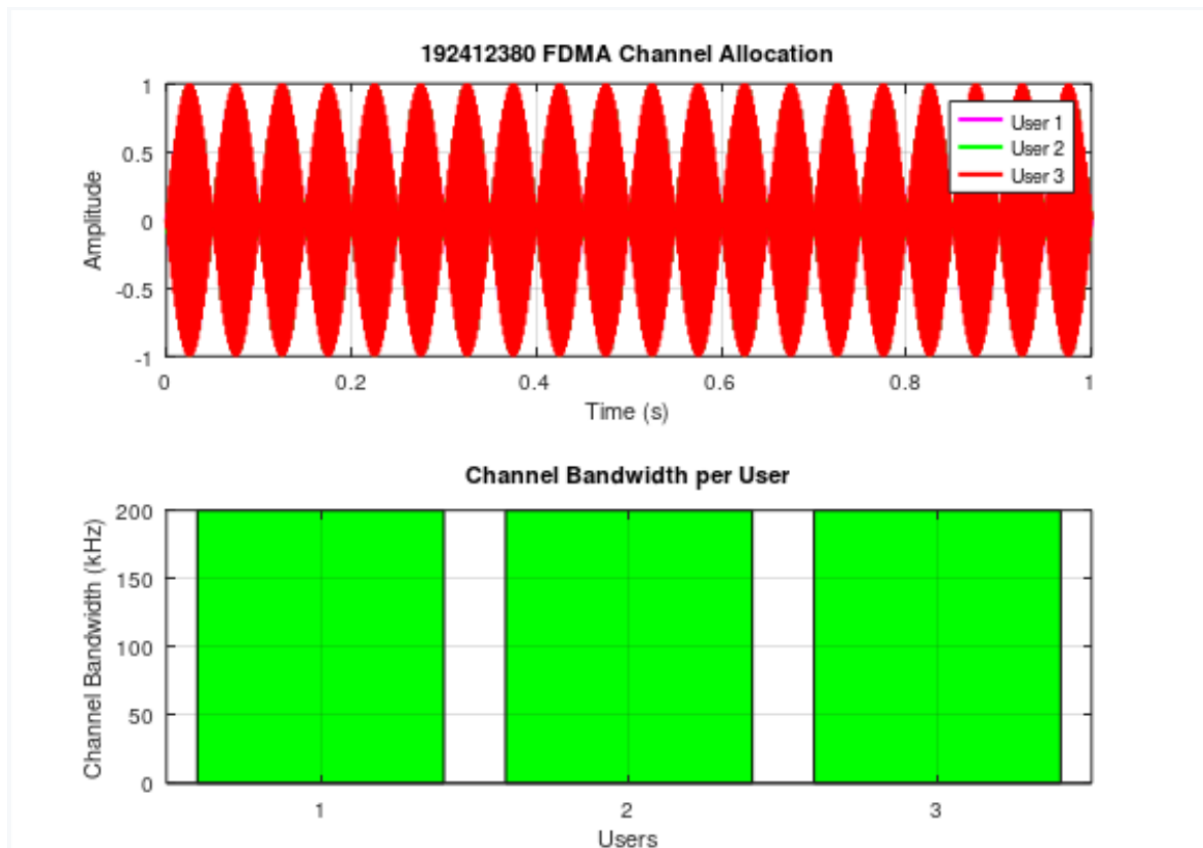
Procedure :

1. Open **MATLAB** and create a new script file.
2. Define the total available bandwidth, number of FDMA users, carrier frequencies, signal power, and noise power.
3. Allocate a unique frequency channel to each user and calculate the **channel bandwidth**.
4. Compute the **data rate** for each user based on the allocated bandwidth and modulation scheme.
5. Calculate the **Signal-to-Noise Ratio (SNR)** using the signal and noise power values.
6. Plot the transmitted signals of all FDMA users using different carrier frequencies.
7. Display the calculated **channel bandwidth, data rate, and SNR** in the MATLAB Command Window.
8. Analyze the results to verify independent channel allocation and evaluate the overall performance of the FDMA system.

Program (Matlab Code)

```
1  clc;
2  clear;
3  close all;
4
5  % Sampling Parameters
6  fs = 1000;
7  t = 0:1/fs:1;
8
9  % FDMA Parameters
10 TBW = 600;
11 Users = 3;
12 CBW = TBW / Users;
13
14 % Carrier Frequencies
15 fc = [100 300 500];
16
17 % Message Signal
18 m = sin(2*pi*10*t);
19
20 % Modulated Signals
21 s1 = m .* cos(2*pi*fc(1)*t);
22 s2 = m .* cos(2*pi*fc(2)*t);
23 s3 = m .* cos(2*pi*fc(3)*t);
24
25 % Plot Modulated Signals
26 subplot(2,1,1);
27 plot(t,s1,'m','LineWidth',1.5);
28 hold on;
29 plot(t,s2,'g','LineWidth',1.5);
30 plot(t,s3,'r','LineWidth',1.5);
31 grid on;
32
33 title('192412380 FDMA Channel Allocation');
34 xlabel('Time (s)');
```

OUTPUT :



Result:

The FDMA system was successfully simulated by allocating separate frequency channels to three users, and the channel bandwidth assigned to each user was determined.

Objective 2:

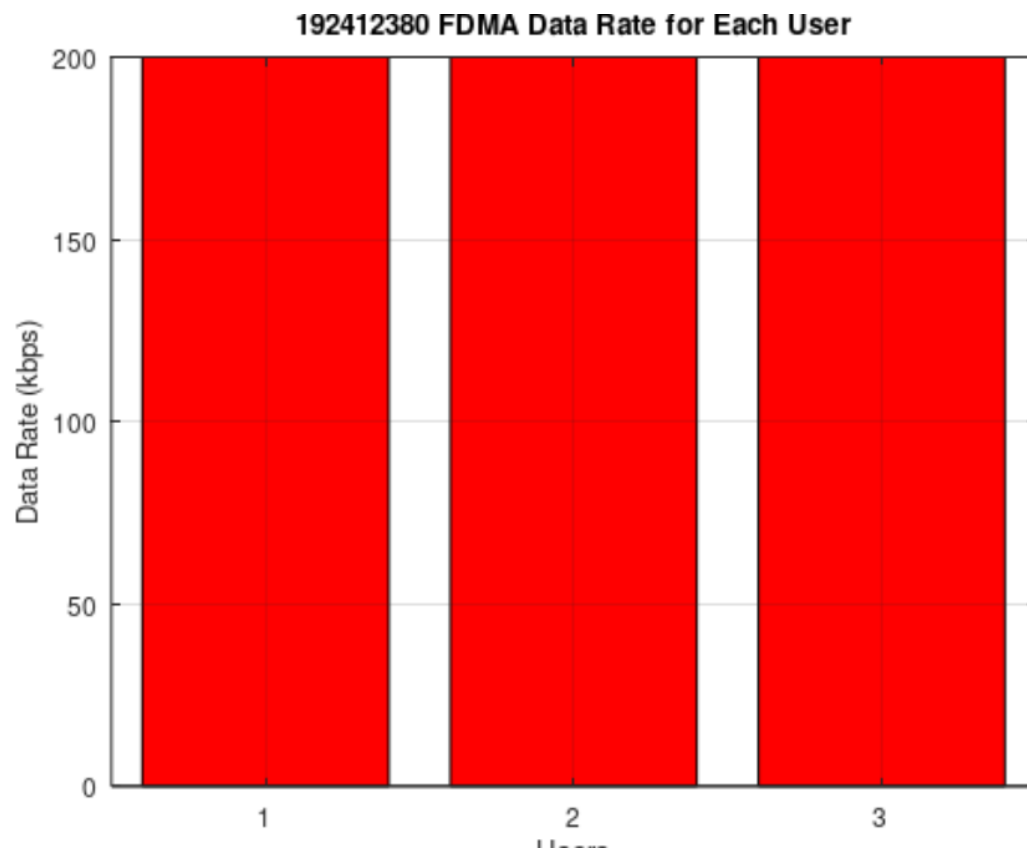
To calculate the data rate achieved by each user in the FDMA system based on the allocated channel bandwidth.

Output Parameter: Data Rate (kbps)

PROGRAM :

```
1  clc;
2  clear;
3  close all;
4
5  % FDMA Data Rate Calculation
6
7  BW = 200;           % Channel Bandwidth (kHz)
8  Users = 3;         % Number of Users
9  M = 2;              % BPSK Modulation
10
11 % Data Rate Calculation
12 DataRate = BW * log2(M);
13
14 % Data Rate for each User
15 Rate = DataRate * ones(1, Users);
16
17 % Display Results
18 fprintf('User\tData Rate (kbps)\n');
19 fprintf('-----\n');
20 for i = 1:Users
21     fprintf('%d\t\t%.2f\n', i, Rate(i));
22 end
23
24 % Plot Graph
25 figure;
26 bar(1:Users, Rate, 'FaceColor', 'r'); % Red bars
27
28 xlabel('Users');
29 ylabel('Data Rate (kbps)');
30 title('192412380 FDMA Data Rate for Each User');
31 grid on;
32 box on;
33
34 set(gca, 'XTick', 1:Users);
```

OUTPUT



RESULT :

The data rate achieved by each user was successfully calculated based on the allocated channel bandwidth, demonstrating equal transmission capacity for all users.

Objective 3:

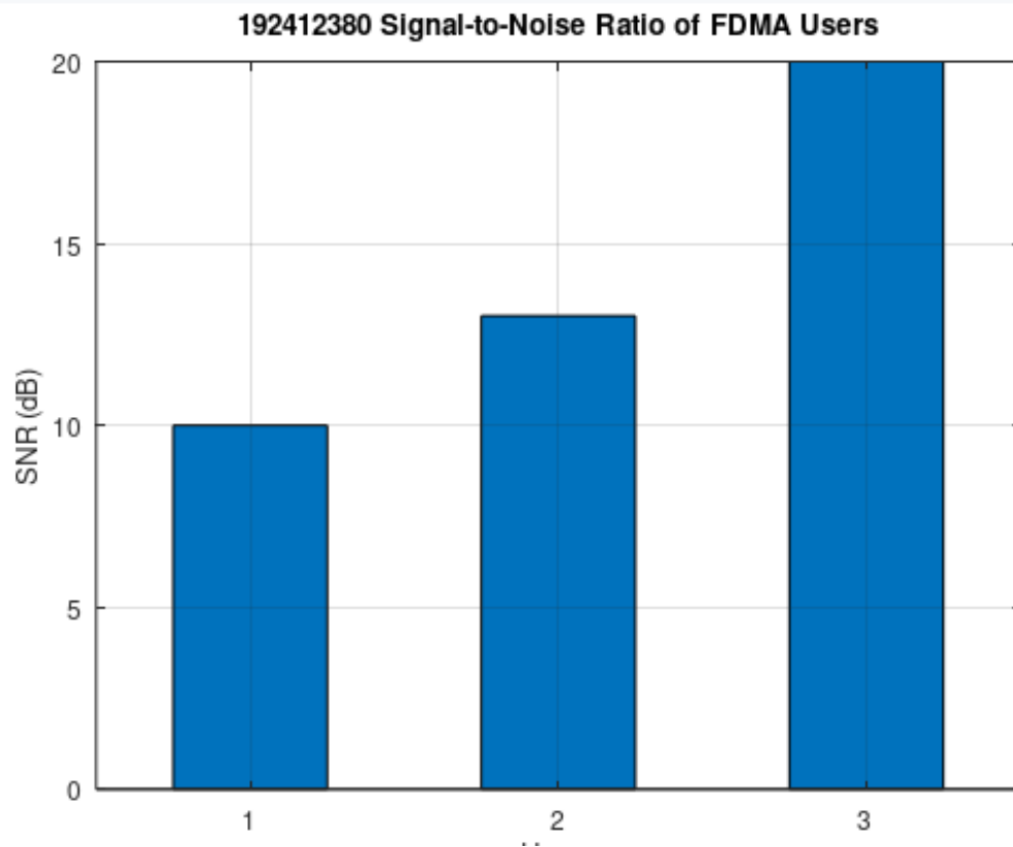
To evaluate the communication quality of the FDMA system by calculating the Signal-to-Noise Ratio (SNR) for different users.

Output Parameter: Signal-to-Noise Ratio (SNR) (dB)

PROGRAM:

```
1  clc;
2  clear;
3  close all;
4
5  % Signal Power (W)
6  Ps = 1;
7
8  % Noise Power (W)
9  Pn = [0.1 0.05 0.01];
10
11 % Users
12 Users = 1:3;
13
14 % Calculate SNR
15 SNR = 10*log10(Ps./Pn);
16
17 % Display Results
18 fprintf('\nUser\tNoise Power(W)\tSNR(dB)\n');
19 fprintf('-----\n');
20 for i = 1:length(Users)
21     fprintf('%d\t%.2f\t%.2f\n',Users(i),Pn(i),SNR(i));
22 end
23
24 % Plot
25 figure('visible','on');
26 bar(Users,SNR,0.5);
27
28 xlabel('Users');
29 ylabel('SNR (dB)');
30 title('192412380 Signal-to-Noise Ratio of FDMA Users');
31
32 grid on;
33 box on;
34
```

OUTPUT:



RESULT:

The Signal-to-Noise Ratio (SNR) for different users was successfully evaluated, showing that higher SNR values provide better communication quality and reliable data transmission.